

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	Slot B	Date:	17.6.26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct (192512274)

I R. Madhumitha R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R. Madhumitha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

1) Effect of changing a Hexadecimal colour values.

→ A hexadecimal colour code is represented colour:

RR = Red

GG = Green

BB = Blue.

Each pair ranges from 00 (minimum) to FF (maximum).

Hex Code	Binary Representation	Displayed Colour.
#FF0000	11111111 00000000 00000000	Red
#F00000	1111 0000 00000000 0000	Dark Red.

2) Effect of Input Values on 8085 Flag Register.

Input	Result	S	Z	AC	P	CY
25H + 15H	3AH	0	0	0	1	0
FFH + 01H	00H	0	1	1	1	1
80H + 80H	00H	0	1	0	1	1

Sign (S): Set if the result is negative

Zero (Z): Set if the result is 00H.

Auxiliary Carry (AC): Set if carry occurs from bit 3 to 4.

Parity (P): Set if the result has even number of 1s.

Carry (CY): Set if carry is generated beyond bit 7.

3) Analysis of Slow System Response.

CPU:

1) Executes instructions for the text editor, browser, and audio player.

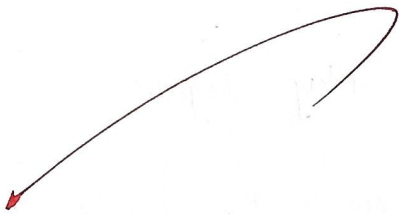
Memory:

1) Stores active program and data.

Input / output:

1) Keyboard and mouse send user input.

2) Hard disk / SSD reads and write files.

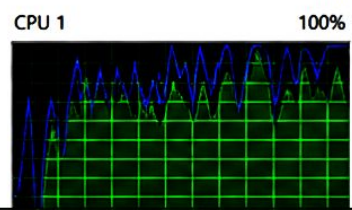
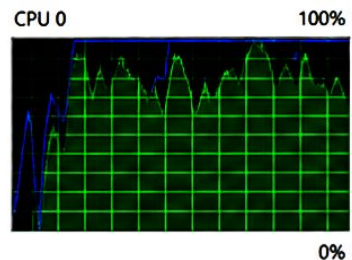
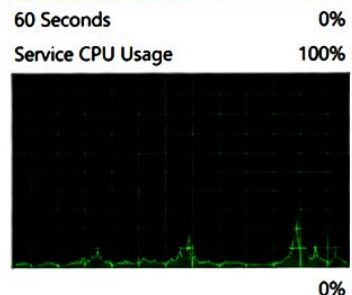
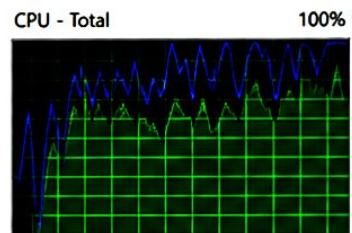


☒ Image	PID	Description	Status	Threads	CPU	Average CPU
☐ MsMpEng.exe	6428	Antimalware Service Executable	Running	145	45	44.32
☐ chrome.exe	12560	Google Chrome	Running	38	15	14.21
☐ chrome.exe	4192	Google Chrome	Running	29	9	8.81
☐ SearchApp.exe	10436	Search application	Running	55	7	6.73
☐ WINWORD.EXE	8840	Microsoft Word	Running	21	5	4.91
☐ chrome.exe	15048	Google Chrome	Running	31	4	3.72
☐ ShellExperienceHost.exe	7720	Windows Shell Experience Host	Running	35	2	2.11
☐ dwm.exe	1180	Desktop Window Manager	Running	17	1	1.23
☐ explorer.exe	5560	Windows Explorer	Running	50	1	0.98
☐ System	4	NT Kernel & System	Running	198	1	0.76
☐ Taskmgr.exe	9764	Task Manager	Running	20	1	0.54
☐ Spotify.exe	13332	Spotify	Running	32	1	0.50
☐ svchost.exe (netsvcs)	2136	Host Process for Windows Service...	Running	39	0	0.32
☐ audiodg.exe	9288	Windows Audio Device Graph Isolatic	Running	12	0	0.22
☐ System Interrupts	-	Deferred Procedure Calls and Inter...	Running	-	0	0.15

Services ■ 10% CPU Usage ▼

Associated Handles 🔍 ↕ ▼

Associated Modules ▼





Registers

A	25
B	15
C	00
D	00
E	00
H	00
L	00
SP	FFFF
PC	0000
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	0
CY	0

Load me at 0000

```

1
2 ; Program to observe flag changes
3
4 MVI A, 25H ; Load first number into Accumulator
5 MVI B, 15H ; Load second number into Register B
6
7 ADD B ; A = A + B (Observe flags)
8
9 SUB B ; A = A - B (Observe flags)
10
11 INR A ; Increment A (Observe flags)
12
13 DCR A ; Decrement A (Observe flags)
14
15 HLT ; Stop execution

```

Decimal - Hex Conversion

Decimal Hex

0 00

⇌ To Hex

⇌ To Dec

I/O Ports

00 - + 00

Ⓒ Update Port Value

Memory

00 - + 00

Ⓒ Update Memory

Start 0000

OK

Address (Hex)	Address	Data
0000	0	3E
0001	1	25
0002	2	06
0003	3	15
0004	4	80
0005	5	90
0006	6	3C
0007	7	3D
0008	8	76
0009	9	00
000A	10	00
000B	11	00

Line No Assembler Message

0 Program assembled successfully

90% Maximum Frequency

Image	PID	Description	Status	Threads	CPU	Average CPU
☐ SearchApp.exe	12348	Search application	Suspended	21	0	0.00
☐ msedgewebview2.exe	13764	Microsoft Edge WebView2	Suspended	45	0	0.00
☐ msedgewebview2.exe	14220	Microsoft Edge WebView2	Suspended	23	0	0.00
☐ msedgewebview2.exe	15832	Microsoft Edge WebView2	Suspended	12	0	0.00
☐ msedgewebview2.exe	16204	Microsoft Edge WebView2	Suspended	17	0	0.00
☐ LockApp.exe	16880	LockApp	Suspended	19	0	0.00
☐ CalculatorApp.exe	4120	Calculator	Suspended	16	0	0.00
☐ SearchApp.exe	17540	Search application	Running	67	7	3.60
☐ System	4	NT Kernel & System	Running	149	3	2.10
☐ System Interrupts	-	Deferred Procedure Calls ...	Running	-	2	1.50
☐ explorer.exe	5288	Windows Explorer	Running	38	1	0.60
☐ dwm.exe	6360	Desktop Window Manager	Running	24	1	0.40
☐ Taskmgr.exe	6892	Task Manager	Running	20	1	0.30
☐ svchost.exe (netsvcs)	7124	Host Process for Windows ...	Running	28	1	0.20
☐ WmiPrvSE.exe	7560	WMI Provider Host	Running	12	0	0.10

Services 6% CPU Usage

Associated Handles  

Associated Modules

